MIL-F-17104E
8 June 1984
SUPERSEDING
MIL-F-17104D
19 February 1971

MILITARY SPECIFICATION

FURNACES, WARM AIR AND HEATERS, UNIT, FORCED AIR CIRCULATION, OIL- AND GAS-FIRED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained, forced-circulation, air heating units designed to distribute warm air for heating purposes either through a duct system for central furnaces or by direct diffusion for unit heaters.

1.2 Classification.

1.2.1 Types, styles, and sizes. The heating units shall be of the following types, styles, and sizes, as specified (see 6.2):

Type I - Central furnace.
Type II - Unit heater.

Style A - Upflow, high-boy.
Style B - Upflow, low-boy.
Style C - Downflow.
Style D - Horizontal flow.

Thermal input as specified in British thermal units per hour (Btu/hr) (see 3.8.1 and 6.2).

1.2.2 Groups. For the purpose of defining requirements, the furnaces and space heaters covered by this specification are classified under the following groups based on the thermal input in Btu/hr.

Group 1 - 22,000 to 225,000 Btu/hr input.
Group 2 - 225,001 to 400,000 Btu/hr input.
Group 3 - Over 400,000 Btu/hr input.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Burner types. Burners furnished on the furnaces and space heaters shall be of the following types, as specified (see 6.2). (Burner types HO, GLO, and GHO shall apply to Group 3 units only.)

Burner type LO - Light-oil-fired (grade 2).
Burner type G - Gas-fired (natural, manufactured, mixed, or liquefied petroleum gas, as specified) (see 6.2).
Burner type HO - Heavy-oil-fired (grade 4).
Burner type GLO - Combination gas- and light-oil-fired.
Burner type GHO - Combination gas- and heavy-oil-fired (grade 4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-F-815 - Fuel Oil, Burner.

MILITARY

MIL-V-173	-	Varnish, Moisture- and Fungus-Resistant (for the Treatment of Communications, Electronic and Associated Electrical Equipment).
MIL-T-704	-	Treatment and Painting of Material.
MIL-B-3180	-	Boilers and Related Equipment, Packaging of.
MIL-T-13867	-	Treatment, Moisture- and Fungus-Resistant for Fire Control, Electrical, and Electronic Instruments and Equipment.
MIL-B-18796	-	Burners, Single Oil, Gas, and Gas-Oil Combination (400,000 Btu/hr and Over Capacity).
MIL-B-18797	-	Burners, Oil Mechanical - Draft, Automatic (Under 400,000 Btu/hr Input Capacity).
MIL-I-24092	-	Insulating Varnish, Electrical, Impregnating, Solvent Containing.
MIL-S-29175	-	Switch, Thermostatic, Low Voltage, Non-Setback/ Setup and Setback/Setup, Temperature Limiting, Heating, Cooling, and Heating-Cooling.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

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MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Department of Energy (DOE)

Title 10, Chapter II, Part 430, Subpart B, Test Procedures.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Federal Trade Commission (FTC)

Title 16 CFR, Part 305. Rules for Using Energy Cost and Consumption Information used in Labeling and Advertising for Consumer Appliances Under the Energy Policy and Conservation Act.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GAS ASSOCIATION LABORATORIES (A.G.A.)

Directory of Certified Appliances and Accessories.

(Application for copies should be addressed to the American Gas Association Laboratories, 8501 East Pleasant Valley Road, Cleveland, Ohio 44131 or 1425 Grande Vista Avenue, Los Angeles, CA 90023.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z21.20 - Automatic Gas Ignition Systems and Components.
Z21.47 - Gas-Fired Central Furnaces (Except Direct Vent and Separated Combustion System Central Furnaces).
Z83.8 - Gas Unit Heaters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems.
ICS 2 - Industrial Control Devices, Controllers, and Assemblies.
ICS 6 - Enclosures for Industrial Controls and Systems.
MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W. Suite 1300, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, Inc., Batterymarch Park, Quincy, Massachusetts 02269.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 296 - Oil Burners.
UL 353 - Limit Controls.
UL 378 - Draft Equipment.
UL 727 - Oil-Fired Central Furnaces.
UL 795 - Commercial-Industrial Gas-Heating Equipment.
UL 900 - Air Filter Units.
- Gas and Oil Equipment Directory.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, Illinois 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The furnaces and unit heaters shall be self-contained, indirect, oil- or gas-fired, forced-air, heating appliances. Type I central

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furnaces shall be designed to supply heated air through a duct system. Type II space heaters may be similar to type I units except that the type II heaters shall be equipped with direct-diffusion, rotatable, sheet metal, louvered nozzles designed to discharge a stream of heated air along a preselected path directly into the space in which the heater is located. Both the furnaces and unit heaters shall consist essentially of a heat exchanger; burner; centrifugal blower(s) or fan(s), as applicable; a sheet metal cabinet-type casing with provisions for duct or discharge-nozzle connection; and all required operating, limit, and safety controls. The furnaces and heaters shall be completely factory-assembled and wired except that the following components may be packed separately at the option of the supplier provided such components accompany the basic unit in shipment: oil burners; blower assembly; discharge nozzles and plenum extensions for type II units; air filter boxes, if furnished; and any other accessories normally shipped disassembled from the unit such as induced draft fan and thermostat.

3.1.1 Upflow furnace. An upflow furnace shall be designed with air flow discharge, vertically upward, at or near the top of the furnace. Upflow, high-boy units shall be arranged with the blower mounted below the heating element. Upflow, low-boy units shall be arranged with the blower mounted beside the heating element. On the low-boy units, both the return air and discharge air connections shall be located on the top of the unit. Type II unit heaters covered by this specification shall be of the upflow, high-boy design only.

3.1.2 Downflow furnace. A downflow furnace shall be designed with air flow essentially in a vertical path wherein the air discharges downward at or near the bottom of the furnace. Downflow furnaces shall be type I, group 1 only and shall be suitable for alcove or closet installation in residential structures.

3.1.3 Horizontal furnace. A horizontal furnace shall be designed basically for low headroom installation with the air flow through the heating element the unit essentially in a horizontal path. Type II units of the horizontal style shall be equipped, when required, with suitable hangers for horizontal mounting.

3.2 Standard commercial product. The units shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the units being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs or brochures, and represents the latest production model.

3.3 First article. When specified (see 6.2), the contractor shall furnish a unit for first article inspection and approval (see 4.2.1 and 6.4).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means

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materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Codes and standards. Forced-circulation furnaces and unit heaters furnished under this specification shall meet the requirements of the following codes and standards to the extent specified:

a. Type I, group 1 and group 2, oil-fired units shall meet the requirements of UL 727 and shall be UL-listed as complete units, including the burners and controls. Burners shall be of the high pressure or low pressure type with direct spark ignition and infrared (cadmium sulfide) primary safety controls in accordance with MIL-B-18797. Varporizing type burners shall not be permitted.
b. Type I, group 1 and group 2, gas-fired units shall meet the requirements of ANSI Z21.47 and shall be A.G.A. Design Certified as complete units, including the burners and controls. Ignition systems shall be of the direct spark, intermittent or interrupted types. Gravity furnaces and continuous pilots shall not be permitted.
c. Type I, group 3 and type II, group 3, oil-fired units shall meet the requirements of UL 727. The oil burners shall conform to MIL-B-18796, and be of the mechanical atomizing type with direct spark or interrupted ignition.
d. Type I, group 3 and type II, group 3 gas-fired units shall meet the requirements of UL 795 or ANSI Z83.8. Heaters with atmospheric burners shall meet ANSI Z83.8 and power burners shall meet MIL-B-18796. Continuous pilots shall not be permitted.
e. Combination oil- and gas-fired units shall comply with the applicable provisions of 3.6c and 3.6d.

3.7 Compliance. Prior to approval of the first shipment, the contractor shall submit for the approval of the contracting officer, or his authorized representative, satisfactory evidence that the furnaces and heaters he proposes to furnish under this specification meets the requirements of UL 727, UL 795, ANSI Z21.47 and ANSI Z83.8 as applicable.

3.7.1 Type I and Type II oil-fired units. Acceptable evidence of meeting the requirements of UL 727 will be the UL listing mark plus a listing in the current UL Gas and Oil Equipment directory indicating that the unit has been approved for the specified installation clearances (see 3.9). Also acceptable is a Certificate of Compliance as part of the first article from a recognized independent testing laboratory acceptable to the Government, indicating the equipment has been tested and conforms to UL 727 and the specified installation clearances (see 3.9).

3.7.2 Type I, group 1 and group 2 gas-fired units. Acceptable evidence of meeting the requirements of ANSI Z21.47 will be the A.G.A. Laboratories

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certification symbol and a listing in the A.G.A. Laboratories Directory of Certified Gas Appliances. Also acceptable is a Certificate of Compliance as part of the first article tests from a recognized independent testing laboratory acceptable to the Government, indicating the furnace design conforms to ANSI Z21.47. The listing or certificate shall indicate that the unit has been approved for the specified input capacity (see 3.8.1), installation clearances (see 3.9), and type of gas (see 1.2.3).

3.7.3 Type I, group 3 and type II, group 3 gas-fired heaters. Acceptable evidence of meeting the requirements of UL 795 will be the UL listing mark or ANSI Z83.8 will be the A.G.A. Laboratories certification symbol and a listing in the A.G.A. Laboratories Directory of Certified Gas Appliances. Also acceptable is a Certificate of Compliance as part of the first article test from a recognized independent testing laboratory acceptable to the Government that the heater has been tested or designed and conforms to UL 795 or ANSI Z83.8, as applicable. The listing or certificate shall indicate that the unit has been approved for the specified input capacity (see 3.8.1), installation clearances (see 3.9), and type of gas (see 1.2.3).

3.8 Performance.

3.8.1 Thermal input. The contractor's rated thermal input in Btu/hr for oil and cubic feet per hour for gas shall be not less than the minimum firing rate as specified (see 6.2). For combination burners, the requirements herein for separate oil-fired and gas-fired burners shall apply.

3.8.2 Efficiency. The thermal efficiency of type I and type II units at the contractor's rated thermal input, shall be not less than the following:

Group 1 - As certified by the contractor to the Federal Trade Commission 10 CFR Part 305, but not less than the 75 percent.
Group 2 and group 3 - 75 percent of the contractor's rated thermal input as listed or certified under the appropriate standards cited in 3.6.

3.8.3 Output capacity. The rated output (bonnet) capacity shall not be less than 75 percent of the contractor's rated thermal input as listed or certified under the appropriate standards cited in 3.6. For group 1, type I furnaces, the applicable information may be provided on the fact sheets required by the Federal Trade Commission.

3.8.4 Temperature rise. The nominal air temperature rise between the inlet and outlet connections at the contractor's rated thermal input shall be 85 degrees Fahrenheit (°F) ± 2°F or other rise is specified (see 6.2). All units shall be capable of being adjusted or regulated to provide a minimum 30°F range of temperature rise, the lower limit being 15°F less than the rated temperature rise and without exceeding a temperature 15°F below high limit cutout of 200°F.

3.8.5 Air flow. The air flow at the rated output (bonnet) capacity, in cubic feet per minute (cfm) of air at the blower or fan intake, shall be established in accordance with UL 727, UL 795, ANSI Z21.47, or ANSI Z83.8 as applicable. The calculated air flow rate for the specified rated temperature rise (85°F) shall be attained against the applicable external static pressure (esp) given in the standards, unless an alternate esp or temperature rise is specified (see 6.2). When an alternate temperature rise and esp are required, the air flow rate shall be established in accordance with the following formula:

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$$\text{Air delivery} = \frac{\text{input} \times \text{efficiency}}{1.08 \times \text{temperature rise}}$$

where: Air delivery = cfm.
Thermal input = manufacturer's rated input Btu/hr (see 3.8.1)
Efficiency = 0.75 (not less than)

$$1.08 = \frac{0.244 \times 60}{13.5}$$

where: 0.244 = specific heat of standard air (70°F db and 50 percent rh)
60 = minutes per hour
13.5 = specific volume of standard air.

3.8.6 Heat transfer rate. The heat transfer rate per square foot of heating surface shall be such that at the specified outlet air test temperature rise, the coincident temperature rise at any point on the primary or secondary heating surface does not exceed the allowable heating element temperature rise specified in UL 727, UL 795, ANSI Z21.47, or ANSI Z83.8 as applicable, for the material used.

3.9 Casing. Each unit shall be enclosed in a sheet metal casing to form a passage around the heat exchanger for the stream of air being heated. The casing thickness shall not be less than the thickness in the reference standards of 3.6. The casing on group 1, vertical units shall enclose the burner and associated fuel system components and controls. On group 2 and group 3 units, the casing shall be braced and reinforced with structural members so that components of the heat exchanger section will not be directly supported solely by the sheet metal casing. Removable panels shall be provided on all units to provide required access to components requiring cleaning, servicing, or adjustment. The interior of the casing shall be equipped with baffles, shields, heat reflecting insulation or inner-liner to prevent the casing heat losses from exceeding 1.5 percent for group 1 and group 2 and two percent for group 3 of the thermal input and to prevent the casing temperature from exceeding the limits neces- sary for compliance with installation clearances allowed under the UL or A.G.A. listings. Unless otherwise specified (see 6.2), units furnished under this specification shall be listed as designed for installation with clearances applicable to the following designations:

a. Group 1, oil-fired:
   1. Styles A, B, and C - UL Form Ia
   2. Style D - UL Form IIb
b. Groups 2 and 3, oil-fired:
   1. Styles A and B - UL Form IIIa
   2. Style C - UL Form III
   3. Style D - UL Form IIIb
c. Group 1, gas-fired: A.G.A. Design Certified for reduced-clearance installations on combustible floors.

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d. Groups 2 and 3, gas-fired: A.G.A. Design certified for standard (6-inch) clearance installation on combustible floors.

3.9.1 Air filters. For group 1 units, integral racks or filter retainers with permanent cleanable or throw-away type type air filters shall be furnished. When specified (see 6.2), air filters for group 2 and group 3 units shall be furnished with integral racks or auxiliary boxes, as specified. Filters may be of the throw-away type or permanent, cleanable type at the option of the supplier unless one type only is specified (see 6.2). Filters shall conform to the requirements of UL 900 for class 1 and class 2. Filters on all units shall be fitted in a manner to preclude air bypassing and to permit ready removal for cleaning or replacement. The total net face area of all filters on each unit shall be such that the face velocity of the air at maximum flow rate attainable within normal operating limits of the unit will not exceed 360 feet per minute (fpm) for standard velocity filters and 520 fpm for high velocity filters.

3.9.2 Warm air connections. Type I units shall include provisions for connecting discharge and return air ductwork. Type II units shall be equipped with directional discharge nozzles designed to rotate 360 degrees. The discharge face of each nozzle of type II units shall be a minimum of 144 square inches and shall be furnished with manually adjustable louvers to permit vertical control of the emitting air stream. The nozzles shall be of the 90-degree pattern unless, for type II, Style D units, 45-degree nozzles are specified (see 6.2). The minimum number of nozzles to be furnished on each space heater shall be two for capacities up to and including size 250,000 Btu/hr, three for capacities 250,000 up to and including 500,000 Btu/hr, and four for capacities over 500,000 Btu/hr.

3.9.3 Automatic vent damper device. When specified (see 6.2), the furnace shall be equipped downstream of the appliance draft hood with an automatic vent damper which is designed to automatically open the venting system when the appliance is in operation and to automatically close off the venting system when the appliance is in a standby condition. The furnace and vent damper combination shall be explicitly designed and certified by ANSI Z21.47, ANSI Z83.8 or listed by UL. An add-on device after installation will not be permitted under this specification.

3.10 Heat exchange surfaces. Heat exchange surfaces shall include the combustion chamber (primary heating surface) plus any secondary surfaces, such as wrap-around radiators, box-type panels, or tube sections, interposed between the combustion chamber and the flue-gas stack connection. On gas-fired units with multiple, manifold-connected burners, combustion chambers may be die-formed integrally with supplemental heat transfer surfaces to provide a unitized, multi-section clam-shell type heat exchanger. All heat exchangers shall be gas-tight construction, designed, and installed to allow the metal to expand and contract within the casing without developing leaks or causing damage to associated parts. Combustion chambers and secondary heating surfaces shall be constructed of materials and sheet metal gages acceptable under UL or ANSI standards for the maximum temperature to which the surfaces

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are exposed at specified test temperatures. Refractory used for chamber linings shall be of the castable, preformed type consisting primarily of alumina, silica, or magnesia compositions with suitable binders. Refractory linings shall be rated to withstand temperatures to 2500°F and shall be replaceable with a minimum of disassembly.

3.11 Burners and fuel systems.

3.11.1 Oil. Group 1 and group 2 oil burners shall be of the high-pressure or low-pressure atomizing type in accordance with MIL-B-18797. group 3 oil burners shall be of the mechanical atomizing type, conforming to MIL-B-18796. Burners shall be designed to fire fuel oil conforming to VV-F-815, grades No. 1 and No. 2 for burner types LO and GLO and grade No. 4 for burner types HO and GHO. The burner shall be completely automatic and shall include a motor, combustion air blower, ignition system, fuel pump, burner tube, and for group 3 units, a safety shutoff valve. Combination burners shall be designed for selective burning either of gas or grade No. 1 or No. 2 fuel oil. Both oil burners and combination burners shall be UL listed.

3.11.2 Gas. group 1 and group 2 gas burners shall meet the requirements of ANSI Z21.47 as specified in 3.6(b). group 3 gas burners shall meet the requirements of UL 795 or ANSI Z21.47 and as specified in 3.6(d). Burners shall be designed, adjusted, rated, and certified to fire natural, manufactured, mixed, or propane gas, as specified (see 1.2.3). Burners shall be rated for use at altitudes to 2,000 feet unless higher altitudes are specified (see 6.2). Gas pressure regulators shall be furnished for all units.

3.12 Ignition systems. The ignition systems shall comply with the pertinent provisions of UL 296, UL 795, ANSI Z21.47, Z21.20, and Z83.8 as applicable. Burners shall be equipped with ignition systems as follows:

a. Ignition systems for group 1 and group 2, gas-fired units, shall be of the direct spark, intermittent or interrupted types. The pilots shall be of the electrically-ignited proven type. Continuous pilots shall not be permitted.
b. Ignition systems for group 3 atmospheric gas burners shall be intermittent or interrupted type in accordance with ANSI Z21.47. Ignition systems for group 3 power gas burners shall meet MIL-B-18796 standards.
c. Ignition systems for group 1 and group 2 oil-fired units shall be of the direct-electrical spark type in accordance with MIL-B-18797.
d. Ignition systems for group 3 oil-fired units shall be direct-electric spark type or interrupted type in accordance with MIL-B-18796.

3.13 Supply fan(s) and blower(s). The blower(s) shall be of centrifugal type and the fan(s) shall be of propeller type, quiet in operation and adequately secured to the shafts. Air flow rating shall be established in accordance with 3.8.5. Bearings shall be self-oiling with adequate oil reservoirs or permanent life-lubricated ball bearings. Oil or grease fittings, when used, shall be located or extended, as required, to provide ready access for periodic lubrication. When the fan or blower is belt driven,

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the motor shall be provided with an adjustable base and guard conforming to OSHA 1910.219 or enclosed in the unit casing. The belt drive shall be in accordance with RMA Engineering standards with service factor of at least 1.2. The direction of rotation shall be clearly and permanently marked on each fan and fan housing. All centrifugal blowers shall be statically and dynamically balanced, and shall be supported by at least two self-alining bearings. Blower speed(s) shall be single, or multi-speed, be adjustable by the use of variable pitch drive sheaves, or be variable speed motors to provide the range of air temperature rises required in 3.8.4, as specified (see 6.2).

3.14 Controls. Each central furnace and unit heater shall be furnished with the following basic operating and limit controls:

a. Means for connecting a thermostatic switch.
b. Blower or fan on-off control.
c. Bonnet temperature limit control.
d. Burner primary safety control.

The blower or fan on-off control and the temperature limit control may be furnished as a single combination control at the option of the contractor. group 3 atmospheric gas burners shall have combustion control systems in accordance with ANSI Z83.8. group 3 powered gas burners, oil burners and combination burners shall be equipped with automatic sequence programmers to effect a timed, pre-ignition, pre-purge period in accordance with MIL-B-18796. Burner combustion control systems shall provide for on-off operation unless as for group 3 units, high-low-off operation or modulated controls are specified (see 6.2). When high-low-off controls are specified, units shall be equipped with multi-position, motorized, fuel-flow control valves, and motorized, two-position damper controls. In lieu thereof, dual high- and low-fire burner nozzles with appropriate automatic control valves may be used for oil; gas valves with three-position, high-low-off actuators may be used for gas. The low position on both high-low-off and modulated controls shall be not less than 33 percent of rated input.

3.14.1 Thermostatic switches. When specified (see 6.2), a room thermostatic switch conforming to MIL-S-29175 shall be furnished with each unit as a separate item for installations and wiring at the site. The thermostatic switch shall be low voltage with temperature limits for family housing of 72°F maximum for heating and 78°F minimum for cooling; or for nonfamily housing of 68°F maximum for heating and 78°F minimum for cooling, as specified (see 6.2). When specified (see 6.2), a thermometer shall be furnished on the thermostatic switch. When specified (see 6.2), alternate or supplementary temperature controls such as system selector switch and fan switch are not required; when heating-cooling switches shall have "emergency heat" position; or multiple setback/setup cycles are desired. The combinations of thermostatic switches available for heating and cooling shall be as in 3.14.1.1 and 3.14.1.2.

3.14.1.1 Group 1 and group 2 units. Thermostatic switches for group 1 and group 2 units shall be one stage heating only; one stage heating and one stage cooling with manual changeover; one stage heating and two stage cooling with manual changeover; or one stage heating and one stage cooling with automatic changeover for nonfamily installations, as specified (see 6.2).

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3.14.1.2 Group 3 units. Thermostatic switches for group 3 units shall be one stage heating only; two stage heating and one stage cooling with manual changeover; one stage heating and one stage cooling with automatic changeover; or two stage heating and two stage cooling with automatic changeover, as specified (see 6.2).

3.14.2 Blower or fan control. The blower or fan control shall prevent operation of the blower or fan after the burner has fired until the discharge air at the bonnet reaches a predetermined temperature. The fan "on" setting shall be adjustable within a minimum range of 90°F to 140°F. The differential setting may be fixed at a point between 24°F and 36°F or shall be adjustable within the minimum range of 20°F to 50°F at the option of the supplier. The blower or fan control shall include a suitable manual switch or setting to permit continuous operation of the fan for summer ventilation purposes. When a temperature sensing blower or fan control is not provided, a device which will provide for blower or fan operation to prevent abnormal air temperature shall be furnished. On group 2 units with high-low-off controls, the blower or fan shall reduce its speed to maintain the normal outlet temperature of the furnace within the limits of 3.8.4.

3.14.3 Limit control. The limit control, conforming to UL 353, shall be designed and installed to shut down the burner before the bonnet air temperature reaches 200°F or at the temperature specified by the applicable standard. The differential shall be fixed at not less than 10°F and not more than 24°F. Controls with adjustable limits up to 250°F shall be equipped with stops factory-set to shut down the burner when the bonnet air temperature reaches 200°F and will require a manual reset.

3.14.4 Primary safety controls Primary safety controls (combustion safeguards) shall be furnished on each oil-, gas-, and combination oil-gas-fired unit. The primary control shall monitor the flame and, in the event of ignition failure on startup or unintentional flame extinguishment during a firing cycle, shall cause safety shutdown of the burner.

3.14.4.1 Group 1 and group 2 units. Primary controls for group 1 and group 2 oil-fired units shall be of the cadmium sulfide type, which responds to the luminous energy in the flame. Controls shall lock out if the flame is not established in accordance with MIL-B-18797 on ignition and shall require manual reset. The automatic ignition and safety shutoff system for group 1 and group 2 gas-fired units shall be of the direct spark, intermittent or interrupted types which shall function to cause complete shutoff of the gas in case of pilot or main flame failure or provide safe re-ignition in accordance with ANSI Z21.47. The timing for actuation of the gas safety shutoff device and automatic valves shall be as specified in ANSI Z21.47.

3.14.4.2 Group 3 units. The flame safeguard and programming control for group 3 units shall be of the electronic type employing flame detectors to monitor the presence of a flame. Flame sensing elements shall be limited to infrared, (lead-sulfide type) or ultraviolet photocells, except that flame rods may be used with gas flames. The programming control, in conjunction with the flame safeguard circuit, shall provide the timed operational sequence in atmospheric gas burners in accordance with ANSI Z83.8 and in power gas burners and oil burners in accordance with MIL-B-18796.

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3.14.5 Oil delay valve. When specified (see 6.2), oil-fired units not equipped with programming controls shall be equipped with a delayed-opening oil shutoff valve. The valve shall automatically delay delivery of oil to the burner until such time as the combustion air fan and, when applicable, the induced draft fan are operating at rated speed. The period of delay from the time the thermostat energizes the burner motor until the fuel valve opens shall not be less than 4 seconds nor more than 8 seconds depending on the motor starting characteristics.

3.15 Electrical requirements Except as otherwise specified herein, all interconnecting wiring for motors, operating controls (thermostatic switches), limit controls, and programmers shall be furnished and installed. When components are shipped unmounted, interconnecting wiring shall be installed up to the location of the unmounted component and the terminals of the wires identified to facilitate connection when the unmounted component is installed. All external wiring shall conform to and be installed in accordance with NFPA Std. No. 70.

3.15.1 Motors. Motors shall be furnished and installed for the fans and blowers, the oil burner when applicable, and the induced draft fan, if furnished. Motors shall be rated and constructed in accordance with the applicable provisions of NEMA MG 1. Motors shall have the electrical characteristics as specified (see 6.2). The blower and fan motor horsepower (hp) shall be sufficient to insure compliance with the limits of motor winding temperature rise specified in UL 727, ANSI Z21.47 or ANSI Z83.8, as applicable, when the units are operating at the rated bonnet capacity and maximum air flow rate against the applicable esp specified in the referenced standards. Unless otherwise specified (see 6.2), motors shall have drip proof enclosures in accordance with NEMA MG 1. Direct drive motors for group 1 units shall not have less than three rated speeds to cover the range of temperature rises specified in 3.8.4.

3.15.2 Magnetic starters. Integral hp motors shall be furnished with magnetic, across-the-line starters rated and constructed in accordance with NEMA ICS 1 and ICS 2. The starter shall provide thermal overload and under-voltage protection and shall include manual and automatic control stations. The overload protection shall include thermal cutouts in all three phases. Unless otherwise specified (see 6.2), enclosures for magnetic starters shall be NEMA type I in accordance with NEMA ICS 6.

3.15.3 Control panel. For group 3 units, the controls, including operating switches, indicating lights, alarms, motor starters, fuses, and circuit elements of the control systems shall be mounted and factory wired on a single control panel or cabinet insofar as practicable in order to centralize the control functions. The control panel shall be mounted on the unit. A schematic wiring diagram of the control circuits shall be furnished in accordance with the referenced standards. Unless otherwise specified (see 6.2), the control panel or cabinet shall be provided with NEMA type 1 enclosure in accordance with NEMA ICS 6. All terminals requiring connection upon installation shall be permanently identified in the instructions and wiring diagrams provided with the equipment. Terminal connections for control functions shall be segregated from power circuit terminals. External wiring shall be in accordance with NFPA No. 70.

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3.16 Accessory equipment.

3.16.1 Induced draft fan. Induced draft fans shall be furnished on group 3 units and, when specified (see 6.2), on group 2 units intended for horizontal, inverted, or other special installations. The fans shall be of the direct-driven or belt-driven centrifugal type. All parts of the fan normally exposed to the flue gases shall be constructed of corrosion-resistant metals or metals suitably treated to resist corrosion.

3.16.2 Vent connections. When specified (see 6.2), the following vent connections and fittings shall be furnished with each type II unit heater:

a. One 90-degree elbow or on tee.
b. Two 45-degree elbows.
c. Ten 2-foot sections of furnace pipe.

The furnace pipe and fittings shall be zinc-coated sheet steel having a nominal thickness of not less than 0.0276 inch (24 gage). The weight of zinc-coating in ounces per square foot shall be not less than 1.25 commercial. Not less than 30 cadmium-plated sheet metal screws for assembling the stack shall be furnished. The nominal size of the furnace pipe and fittings for the specified furnace sizes shall not be less than shown in table I.

TABLE I. Furnace pipe sizes for type II space heaters.

Rated input Btu/hr thousands	Size inches w/o exhauster	w/exhauster
22 - 225	8	6
226 - 400	10	8
401 - 1250	NA	10
1300 - 2500	NA	12
2600 - 3125	NA	128 sq. inches
3200 - 4000	NA	182 sq. inches

Note: If furnace piping and fittings are specified in the contract or order and the standard vent connection on the furnace is other than the size specified for the furnace pipe, a suitable adapter shall be furnished.

3.16.3 Draft regulator. When specified (see 6.2), a barometric draft regulator shall be furnished for oil-fired units. The regulator shall be adjustable to encompass a range of settings from 0.02 to 0.08 of an inch of water and shall meet the requirements of UL 378. When both vent connections and a draft regulator are specified in the contract, the draft regulator shall be mounted in a tee-section. Otherwise, the regulator shall be furnished separately for mounting at the site.

3.17 Treatment and painting. Unless otherwise specified (see 6.2), all parts of the equipment normally painted, shall be treated and painted in accordance with the manufacturer's standard practice. When specified (see 6.2), treatment and painting shall be type A of MIL-T-704 and the color green shall be No. 24087 of FED-STD-595.

3.18 Electromagnetic interference suppression. When specified (see 6.2), motors and controls shall be designed and equipped to meet class C3, group I of MIL-STD-461.

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3.19 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be provided in accordance with the manufacturer's standard practice. Parts requiring lubrication shall be so located as to make the lubricating points easily visible and accessible. All parts requiring lubrication should be properly lubricated before delivery.

3.20 Fungus resistance. When specified (see 6.2), electric components and circuit elements including terminal and circuit connections shall be treated for moisture and fungus resistance in accordance with MIL-T-13867 as specified herein. Electrical components with temperature rise of 40° centigrade (C) or less shall be coated in accordance with type II of MIL-T-13867 with fungus resistance varnish conforming to MIL-V-173. Components with a temperature rise exceeding 40°C shall be coated in accordance with type III or IV of MIL-T-13867 with varnish conforming to MIL-I-24092 with grade and class of varnish dependent on the component, class of insulation above 40°C and the recommendation of the manufacturer of the equipment.

3.21 Identification marking. Identification shall be permanently and legibly marked directly on the furnace or on a corrosion-resisting plate, firmly affixed to the item. The information shall include the manufacturer's name or trademark, model number, and serial number, in accordance with UL 727, UL 795, ANSI Z21.47, or ANSI Z83.8, as applicable.

3.22 Instruction plates. The equipment shall be provided with instruction plates describing special or important procedures for operating and servicing equipment and warnings of hazardous procedures. Instruction plates shall comply with the provisions of ANSI Z21.47, ANSI Z83.8, UL 727 or UL 795, as applicable.

3.22.1 Military marking. When specified (see 6.2), units shall be marked for identification in accordance with MIL-STD-130 in addition to the marking specified in 3.21. The marking shall be applied to the units on a plate which shall be made of a material which will last and remain legible for the life of the equipment. Plates shall be attached to the equipment by bolts, screws, or rivets in a conspicuous location affording protection against damage or disfigurement during normal use of the equipment.

3.23 Technical manuals. The contractor shall provide technical manual(s) conforming to the Data Item Description(s) listed on the Contract Data Requirements List (DD Form 1423). Each manual shall contain at least the basic operating and installation instructions required under UL 727, UL 795, ANSI Z21.47, or ANSI Z83.8, as applicable.

3.24 Workmanship.

3.24.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.24.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with commercial practice, and all bolts, nuts, and screws shall be tight.

3.24.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.24.4 Machine work. Tolerances and gages for metal fits shall conform to the standards of commercial practice. Finished contact and bearing surfaces shall be true and exact.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one of each classification of furnace as specified when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4 and 4.6, and, when specified, the preproduction pack inspection of 4.7 (see 4.7 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.7.

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4.3 Examination. The units shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.3.1 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standards cited in 3.6.

4.4 First article tests.

4.4.1 Operational test. The first article shall be subjected to an operational fire test to verify the functional adequacy of burners, blowers, and controls. This unit shall be set up with the test duct as specified in the applicable UL or ANSI standard to maintain the required external static test pressure. The unit shall be operated sufficiently to assure equilibrium is established with the air flow rate adjusted to give the rated outlet temperature of 85°F rise or as required in the contract. Under these conditions, the unit shall operate continuously at rated thermal input without interference from the limit control. After the initial test period, the limit control cutout point shall be checked as prescribed in UL 727, UL 795, ANSI Z21.47, or ANSI Z83.8. During the operational test, the ignition system shall be checked for proper light-off, the burner flame observed for proper shape, color, or excessive smoking, and the blower control checked for proper setting.

4.4.2 Flame safeguard test. After completion of the test of 4.4.1, the flame safeguard control on the sample test unit shall be tested by manually closing the main fuel supply valve to simulate a flame failure. The time required for safety shutdown of the unit shall be checked for requirements of 3.14.4.

4.4.3 Heat exchanger tightness test. During or after the test of 4.4.1, the heat exchanger on the sample test unit shall be tested for tightness. The test may be conducted by introducing a fuming or smoking material into the combustion chamber; by taking comparative carbon dioxide readings at two different external static pressures; by pneumatic pressure tests conducted prior to final assembly of the units; or by an equivalent method normally used by the manufacturer.

4.4.4 Thermal efficiency. The unit shall be tested to verify compliance with the thermal input of 3.8.1 and the efficiency requirements of 3.8.2. During the tests, the stack temperature and smoke reading shall also be recorded. Tests shall be conducted as follows:

a. For group I units; in accordance with DOE 10 CFR, Part 430, Subpart B, Appendix N for steady state efficiency only.
b. For group II, group III, and type 1 units, in accordance with UL 727, UL 795, ANSI Z21.47, and ANSI Z83.8, as applicable.

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Efficiency tests shall be conducted only after satisfactory completion of the tests of 4.4.1, 4.4.2, and 4.4.3.

4.4.5 Control tests. In addition to the tests of 4.4, the following controls on group 3 first article shall be checked, as applicable, to verify compliance with the applicable requirements of this specification and of referenced standards:

a. Programming sequence control.
b. High-low-off control.
c. On-off control.

4.5 Quality conformance tests. The contractor shall have an established test system which shall assure compliance with the requirements of the specification and of the referenced standards. The system shall include the methods, procedures, controls, records, and maintenance of the system to provide verification for the certifying agency of compliance with the requirements of this specification and of the referenced standards and has been found mutually acceptable to the contractor and the certifying agency.

4.6 Test for electromagnetic interference suppression. When conformance to electromagnetic interference suppression limits is required, the equipment shall be tested in accordance with MIL-STD-462 to determine conformance to the electromagnetic suppression characteristics of 3.18. When suppressed to conform to 3.18 the contractor may, upon approval of the contracting officer, furnish a certificate of compliance in lieu of the test, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed. This requirement shall apply to the first production unit when a first article is not required.

4.7 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be in accordance with MIL-B-3180 except that type P-21 preservative shall be used in lieu of P-3 preservative. The level of preservation and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. Central furnaces are intended for installation in family housing or nonfamily housing structures primarily to supply heated air through a sheet metal duct system. Unit heaters are intended primarily for installation in such nonfamily structures as warehouses, industrial shops, auditoriums, hangars, and gymnasiums where warm air will be directly diffused into the space in which the heater is located. Downflow furnaces can be used

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to an advantage with under-floor duct systems in structures having no basements for central-furnace location. Downflow furnaces are also referred to as counterflow units since the heated-air path runs in a direction counter to the flue gas passage.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, style, and burner type required; type of gas for burner types G, GLO, and GHO (see 1.2.3 and 3.8.1).
c. When a first article is required (see 3.3).
d. Thermal input required Btu/hr (see 3.8.1).
e. When the nominal air temperature rise shall be other than 85°F (see 3.8.4).
f. When external static pressure and air flow (cfm) other than rated shall apply (see 3.8.5).
g. When installation clearances shall be other than as specified (see 3.9).
h. When group 2 and group 3 units shall be equipped with air filters and, if so, when a particular type is required (see 3.9.1).
i. When discharge nozzles for suspended, horizontal, space heaters shall be of the 45-degree pattern (see 3.9.2).
j. When automatic vent damper device is required (see 3.9.3).
k. Heating value and supply pressure for gas-fired units if other than natural gas will be used; altitude at which gas-fired units will operate if above 2,000 feet (see 3.11.2).
l. When blower speeds on units shall be adjustable and with number of speeds (see 3.13).
m. When group 3 units shall be equipped with high-low-off controls or modulated controls (see 3.14).
n. When thermostatic switch is required; whether for family housing, or nonfamily housing; when thermometer is required; or whether alternate or supplementary controls are required (see 3.14.1).
o. When thermostatic switches for group 1 and group 2 units require heating only, heating-cooling, one stage, two stages and manual or automatic changeover (see 3.14.1.1).
p. When thermostatic switches for group 3 units require heating only, heating-cooling, one stage or two stage, and manual or automatic changeover (see 3.14.1.2).
q. When an oil delay valve is required on group 1 and group 2 units (see 3.14.5).
r. Required electrical characteristics of motors; when other than drip proof motors are required (see 3.15.1).
s. When other than NEMA type 1 enclosures are required for magnetic starters (see 3.15.2).
t. For group 3 units, when other than NEMA type 1 enclosures are required for the control panel (see 3.15.3).
u. When induced draft fans shall be furnished on group 2 units (see 3.16.1).
v. When vent connections are required for space heaters (see 3.16.2).
w. When a draft regulator is required (see 3.16.3).
x. When treatment and painting procedures shall conform to type A of MIL-T-704 (see 3.17).

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y. When electromagnetic interference control is required (see 3.18).
z. When lubrication shall be different (see 3.19).
aa. When treatment for fungus resistance is required (see 3.20).
bb. When Military marking in accordance with MIL-STD-130 is required and the information to be marked (see 3.22.1).
cc. When a preproduction pack inspection is required and time frame required for submittal (see 4.7).
dd. Level of preservation and level of packing required (see 5.1).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete unit or it may be a standard production item from the contractor's current inventory. The contracting officer should include specific instructions, tests, and approval of the first article. The first article should only be required when compliance with the applicable standards cited in 3.6 or certification specified in 3.7 are not made available.

6.5 Cross-reference of classification. The following cross-reference to input capacity classification applies.

MIL-F-17104D	MIL-F-17104E
Sizes - 40 through 1,000 (see Table I)	Input capacity as specified in Btu/hr (see 3.8.1 and 6.2)

6.6 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing Activity:

Navy - YD

(Project 4520-0272)

Review Activities:

Army - MD
DLA - CS

User Activity:

Army - CE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-17104E	FURNACES, WARM AIR AND HEATERS, UNIT, FORCED AIR CIRCULATION, OIL- AND GAS-FIRED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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